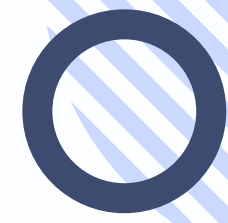


FootSense

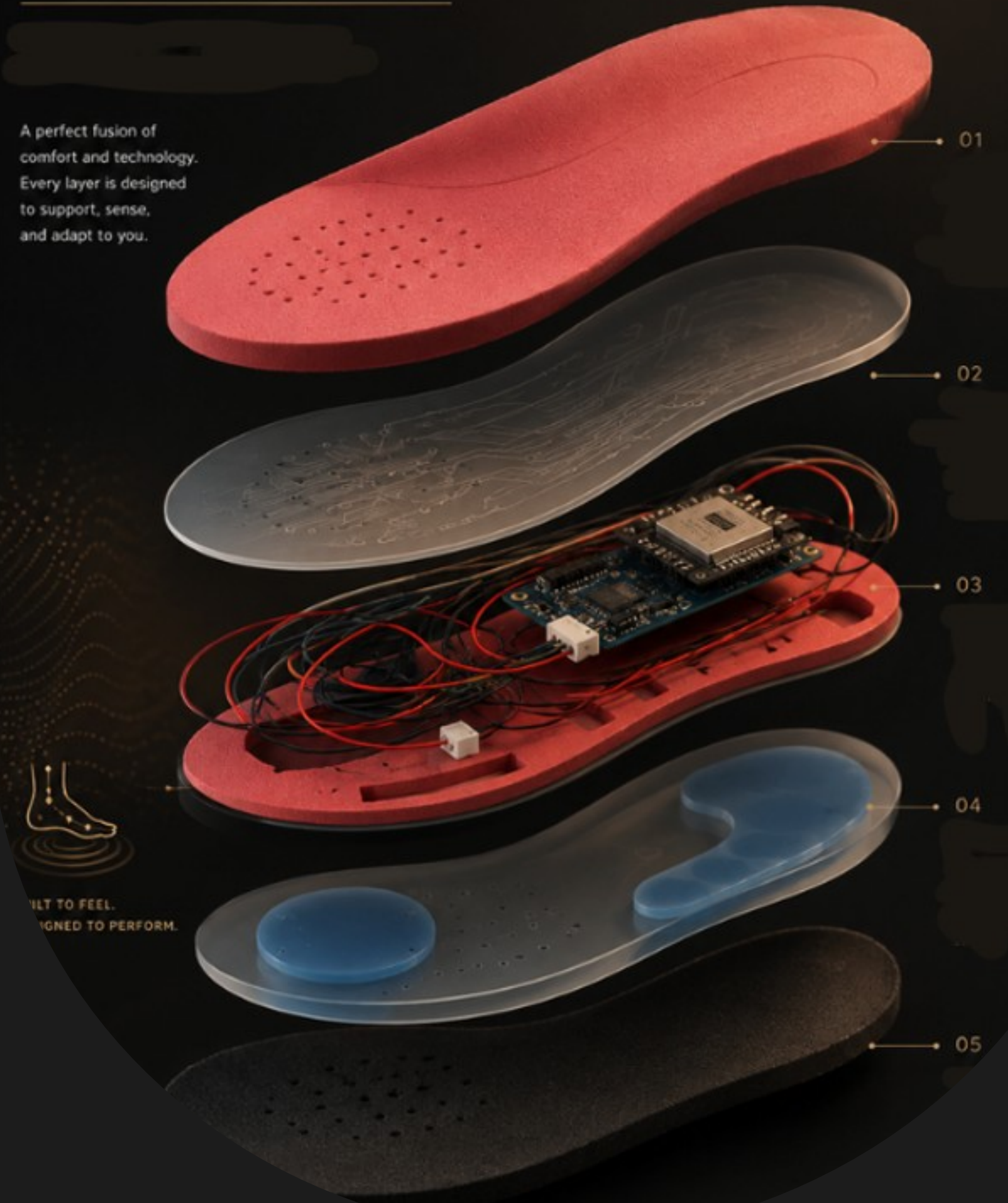
smart insole tracker

A 10 crore rupee problem,
solved with a soldering iron

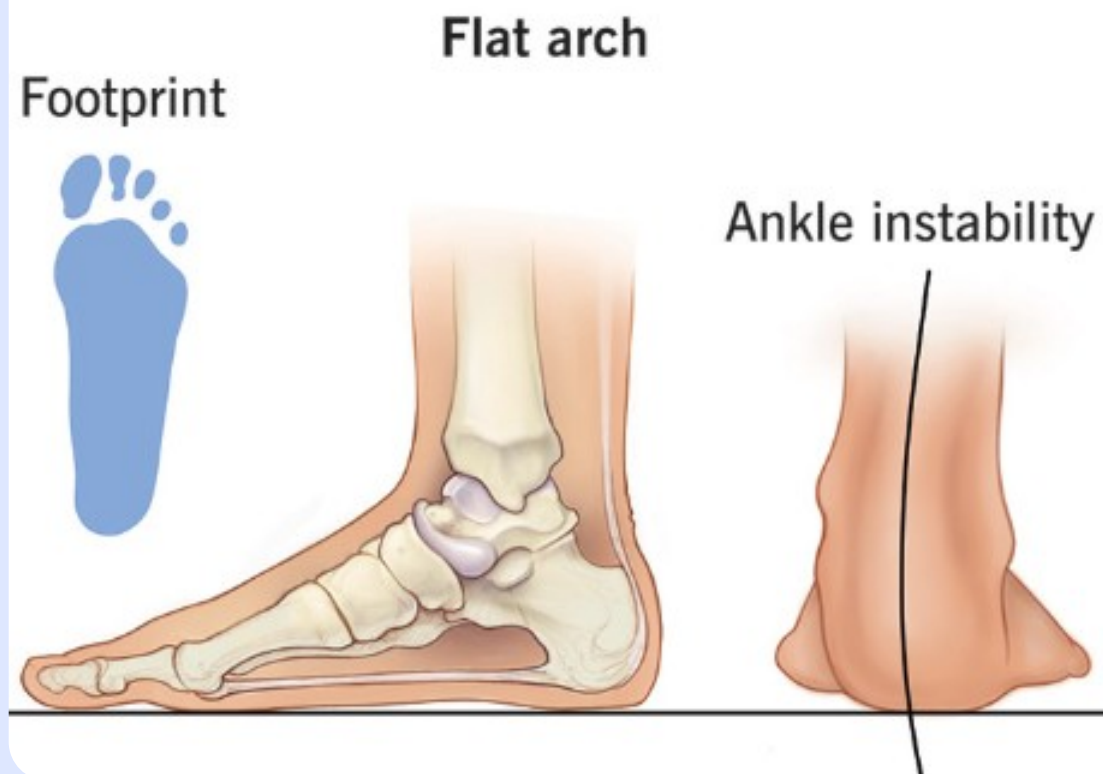
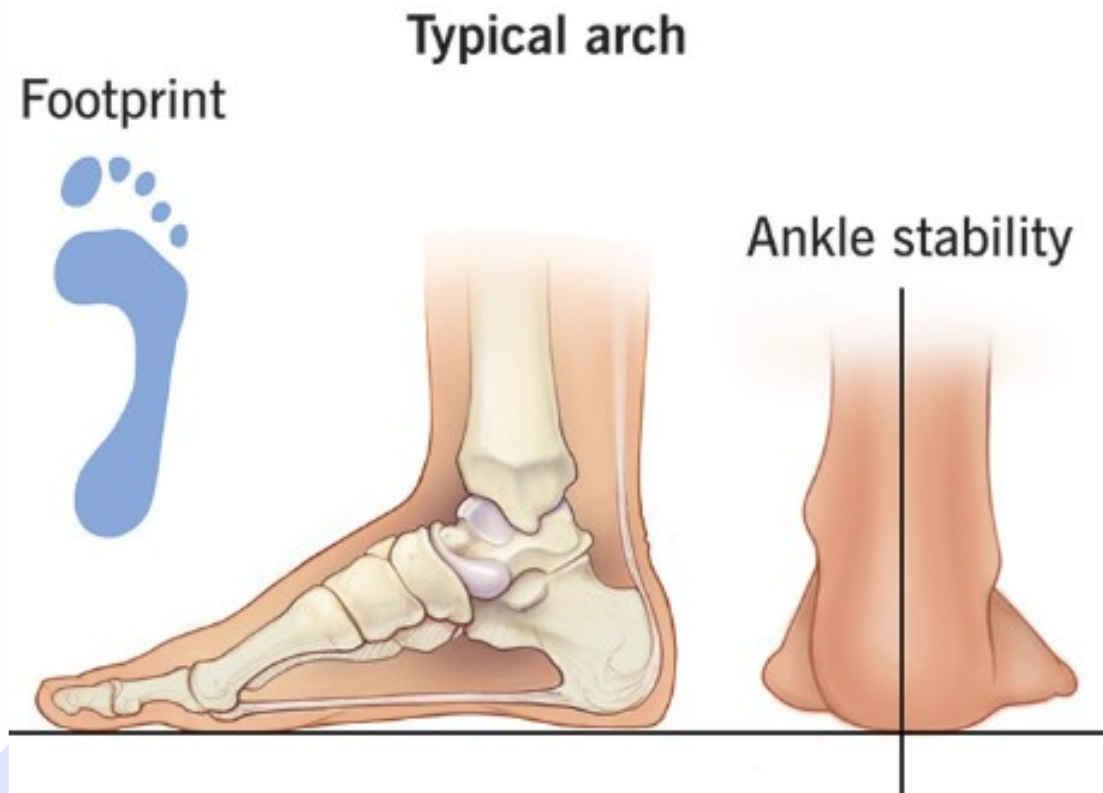


SMART INSOLE

A perfect fusion of
comfort and technology.
Every layer is designed
to support, sense,
and adapt to you.



Flat feet (pes planus)



The Problem

- ✓ 9–18% of Indians show flat feet, depending on which clinical test you use
- ✓ One study of traffic police in Gujarat found flat feet in 33–34% of officers — mostly from standing all day
- ✓ 24.8% prevalence found in a Delhi hospital screening study



The Expensive Way

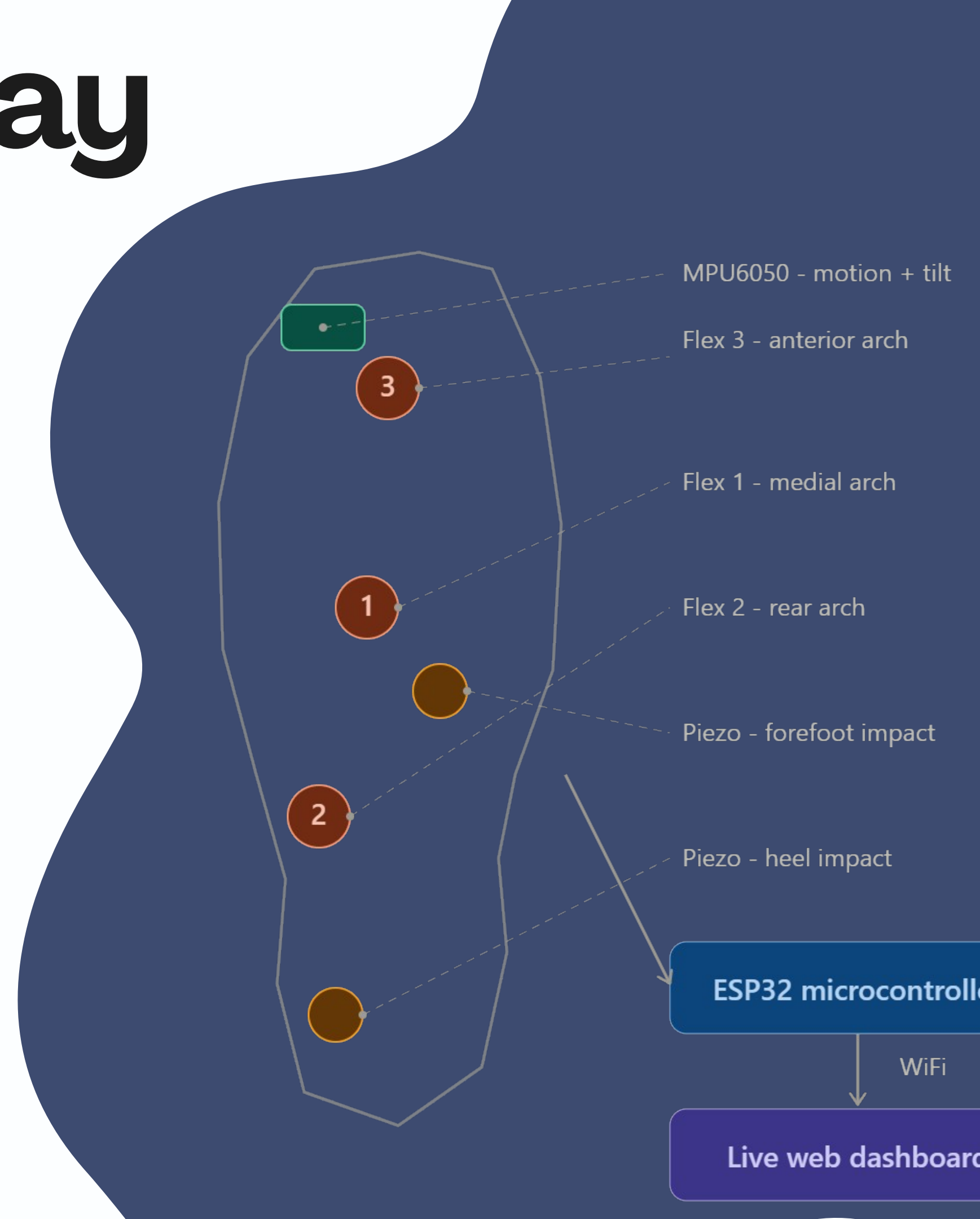
Clinical Gait Analysis System

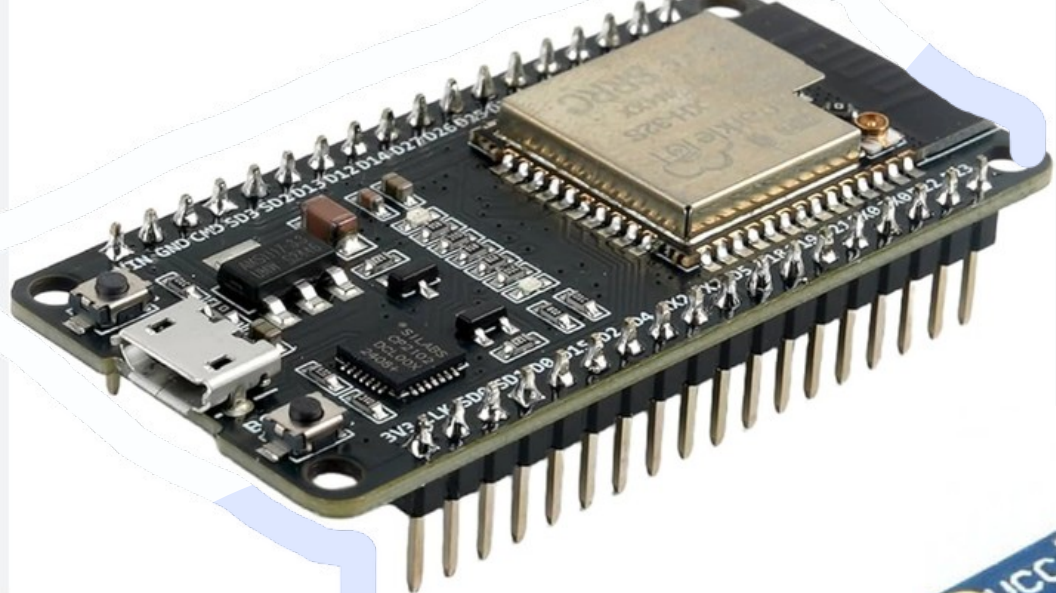
VS

FootSense

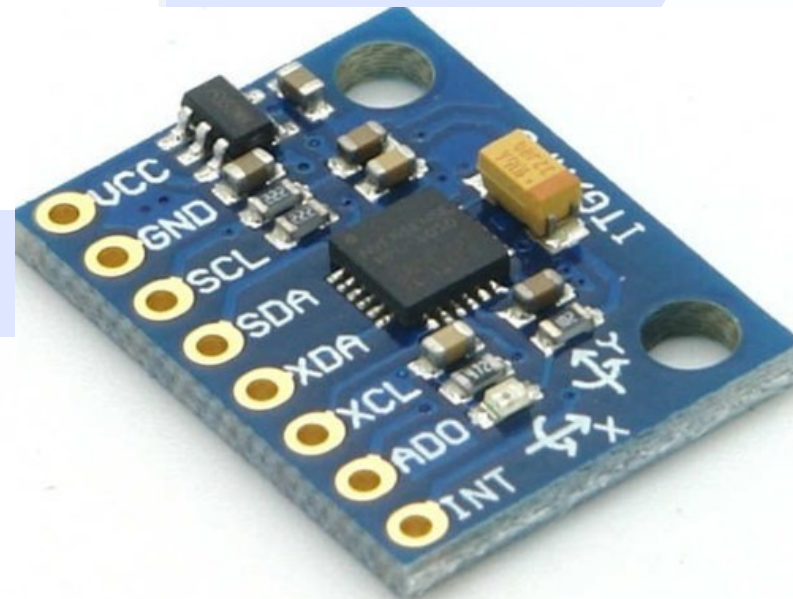
3D Gait Analysis System: ₹10,00,000+

FootSense: a few hundred rupees in sensors and
a Sunday at Maker's Asylum





ESP32



MPU6050



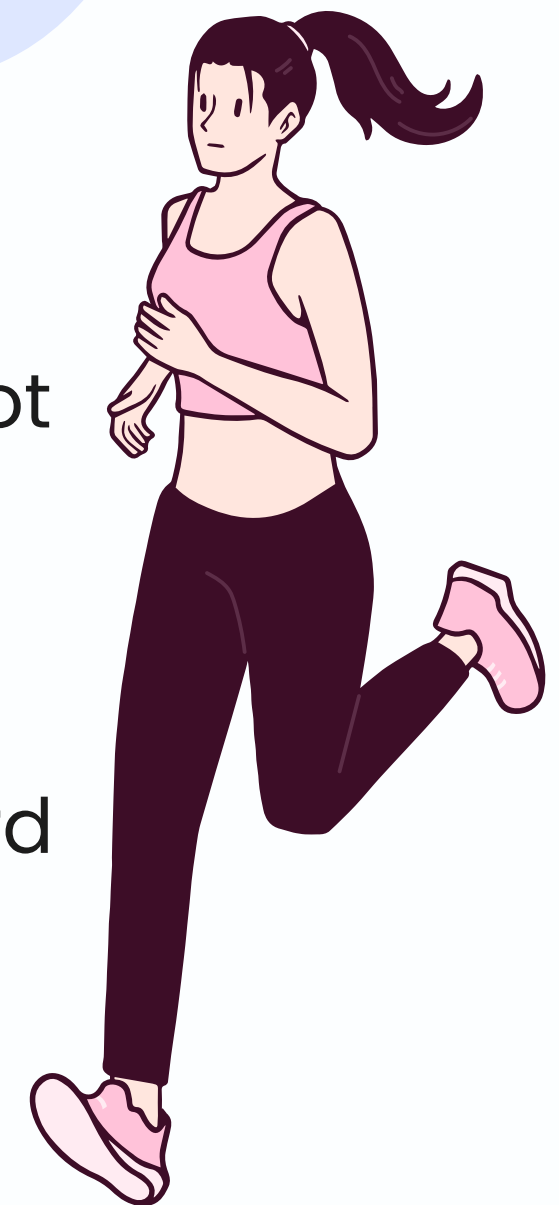
flex sensor



piezo disc

What's Actually in the Shoe

- 3 flex sensors → arch shape (flat / normal / high)
- 2 piezo discs → heel & forefoot impact
- MPU6050 → step count & impact angle
- ESP32 → WiFi → live dashboard

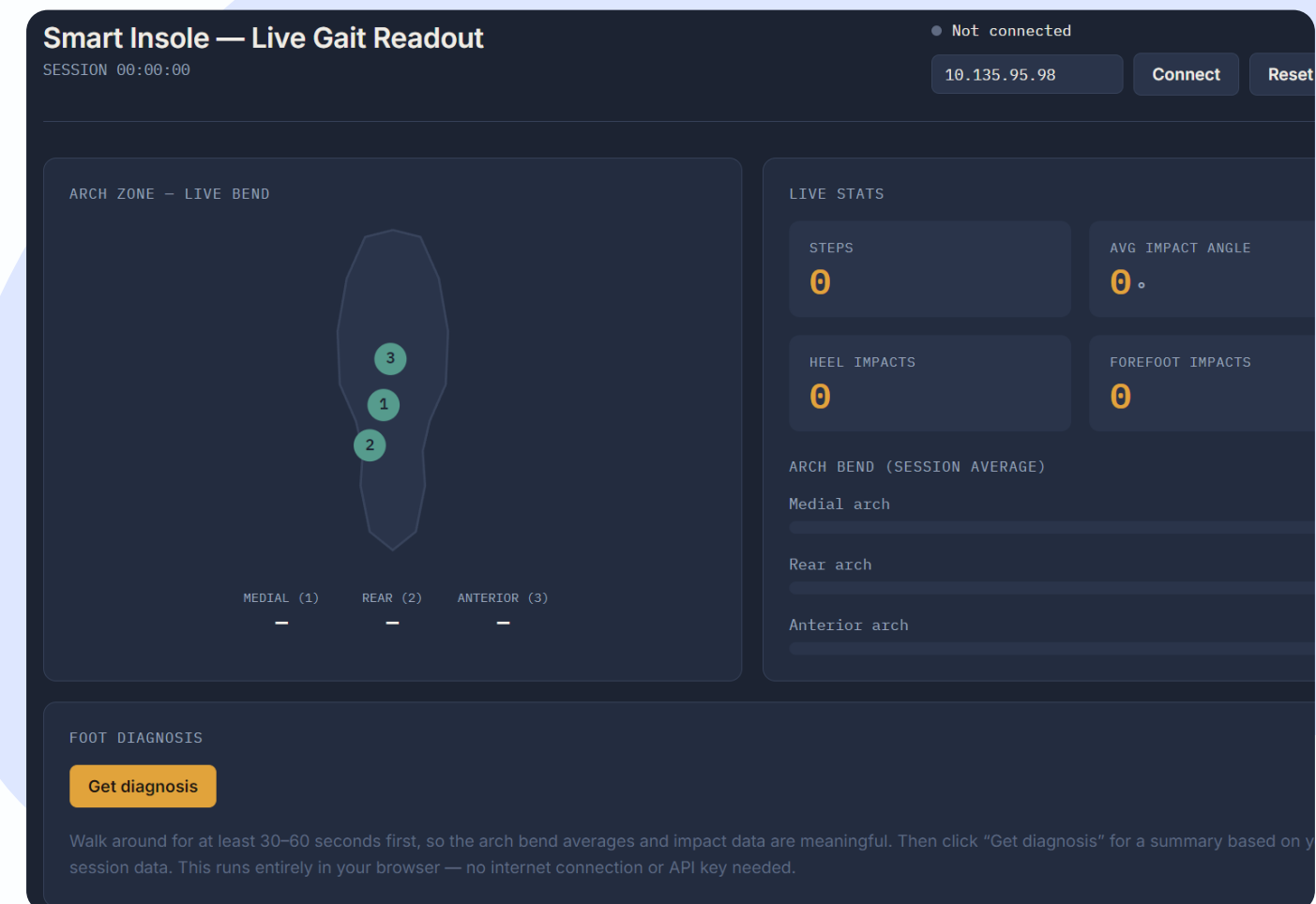
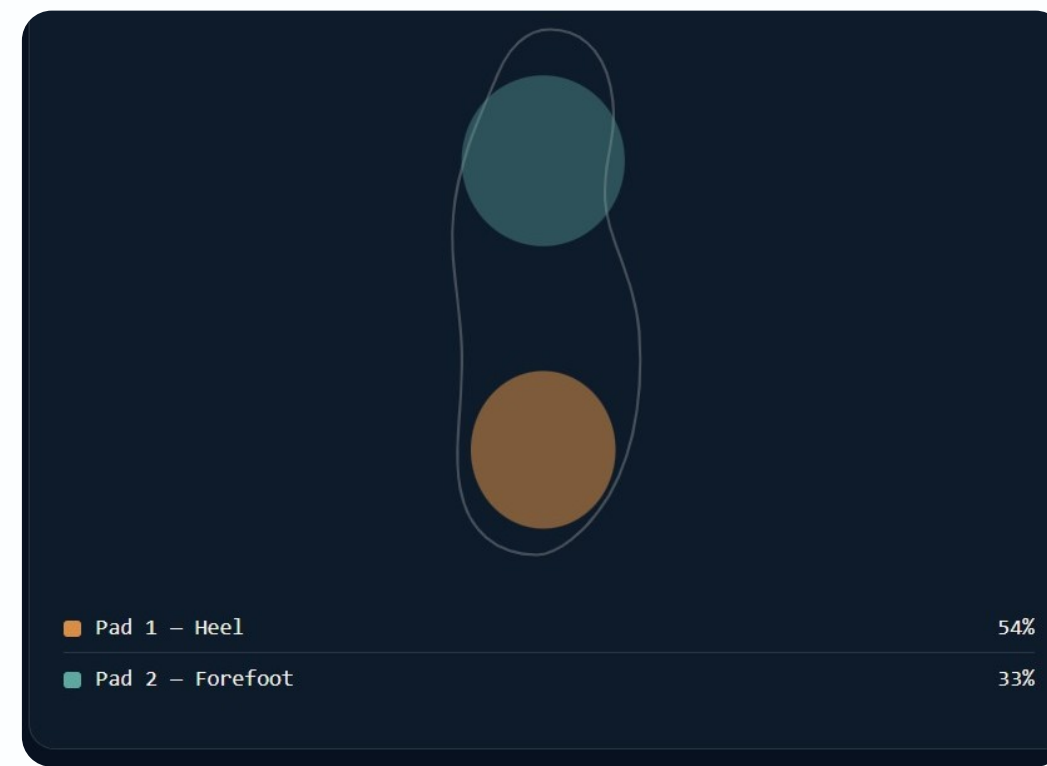


It Talks Back

Live Dashboard



- ✓ Real Time Arch Bend
- ✓ Foot Diagnosis
- ✓ Impact Pressure (Graph and Data)
- ✓ Step Count + Impact Angle

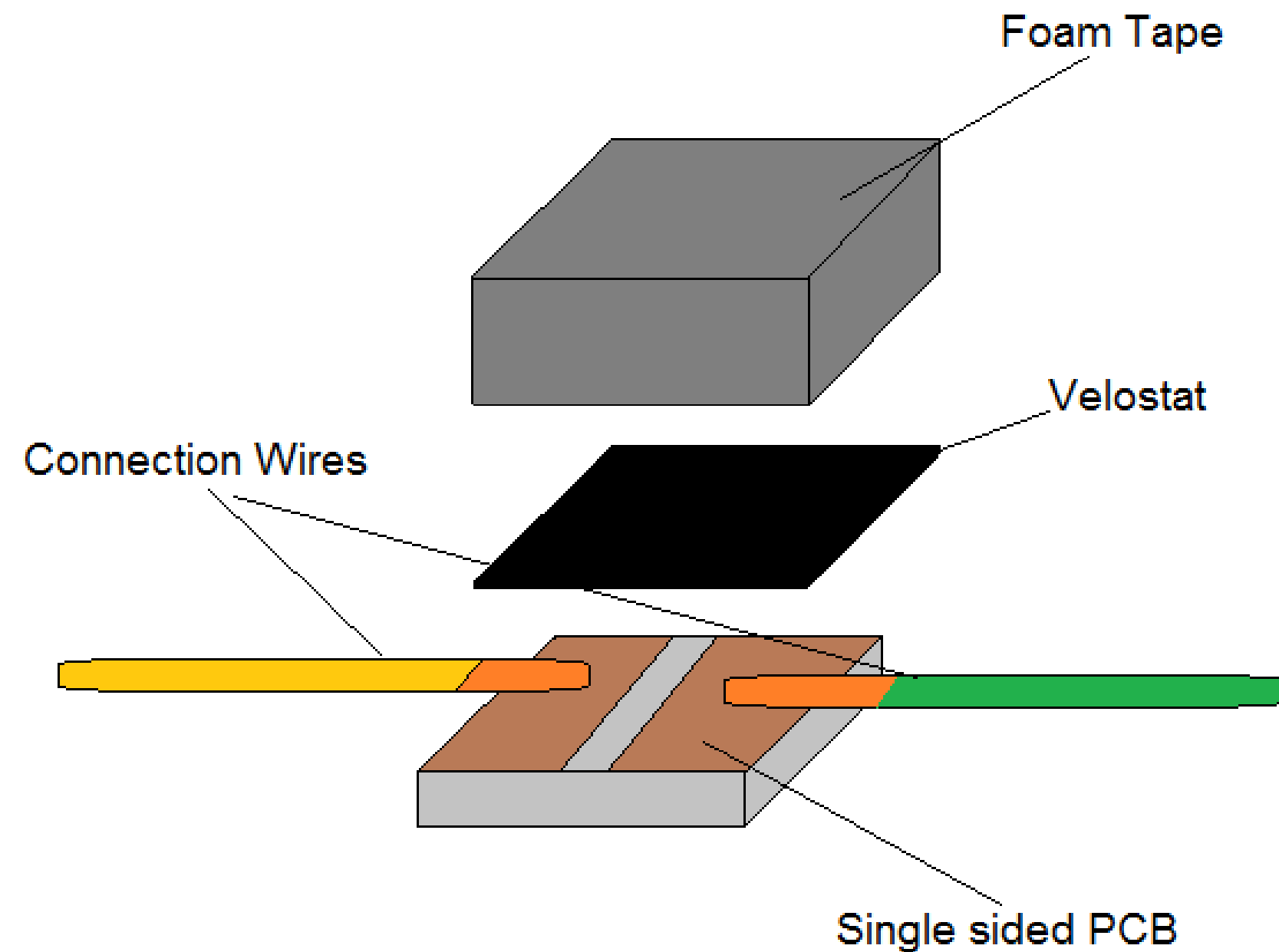




The Truth

- Load cell + 4-sensor bridge: didn't work, never fully diagnosed why
- Switched to flex sensors + piezo discs/velostat instead
- Chased ghost GPIO pins across three different boards
- Eventually got a working ESP32 + WiFi + live dashboard pipeline

The Velostat Version

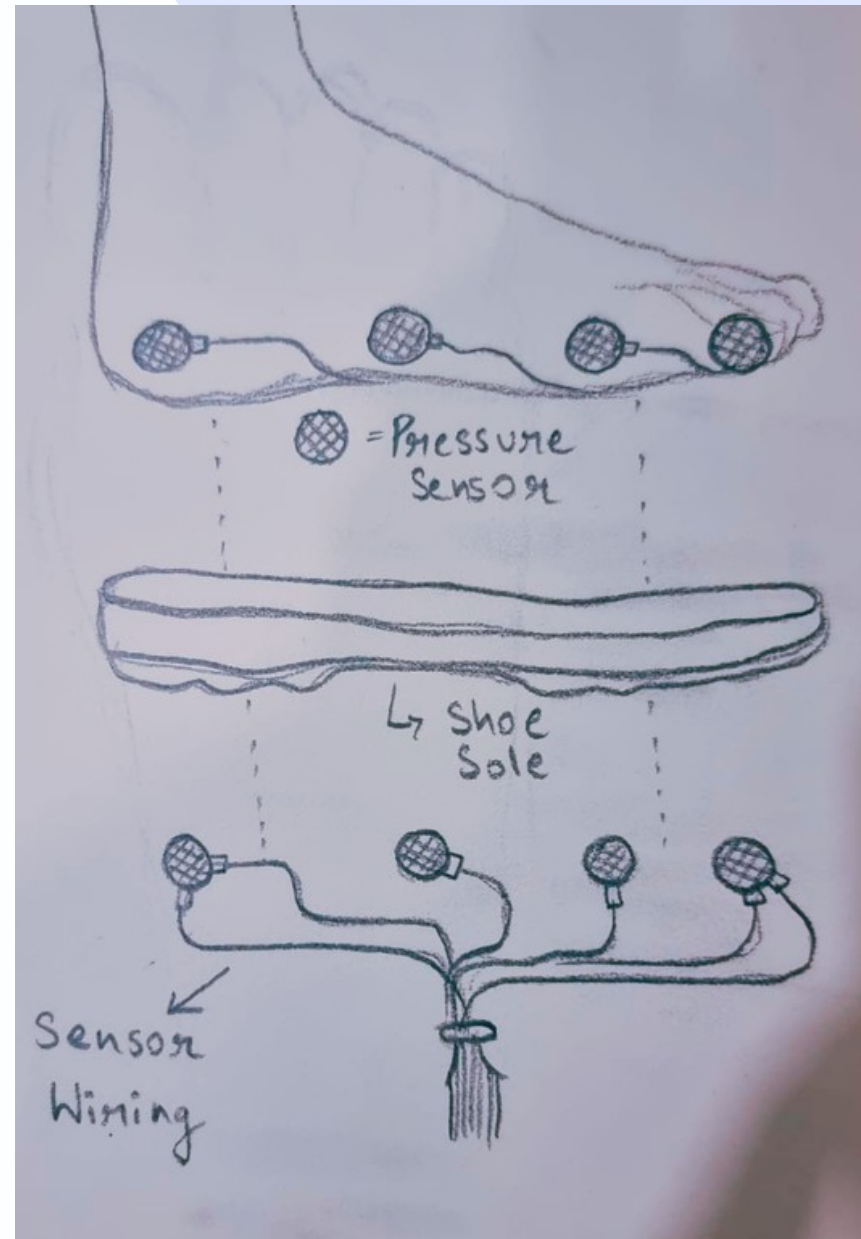


- Pressure-sensitive sheet, not discrete sensors
- Resistance drops the harder you press — wired as a simple voltage divider
- Gives a continuous pressure map instead of fixed points
- Closer to what ₹10,00,000 clinical mats actually do
- Tradeoff: more finicky to calibrate consistently
- Built in parallel — not a replacement, a second real path

The Truth Again

Honest Limitations

- This is a relative signal, not a medical diagnosis
- Tested on one person, on one foot.



Roadmap



✓ Q1 Development

Finalize innovative product design and successfully complete precise sensor calibration process. Design a finalised protected prototype and apply for a license,



✓ Q2 Testing

Testing the product and collecting data. Doing further extensive research and training a neural network to develop an AI/ML finished software. Connect with doctors, share the data and use their feedback.

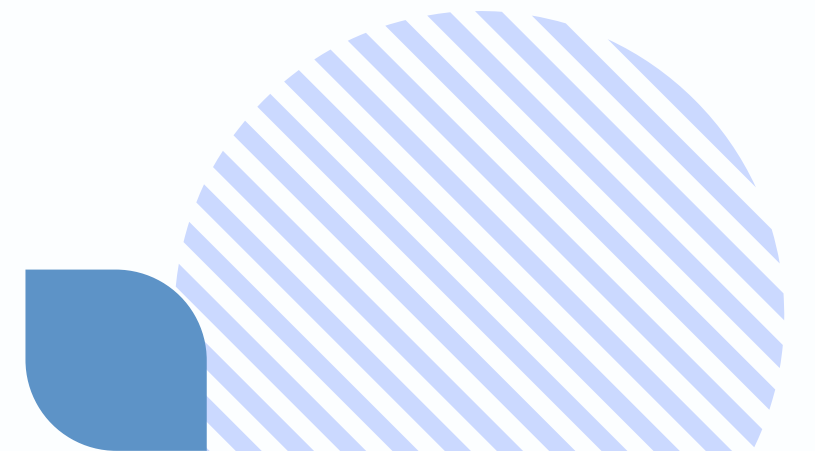
✓ Q3 Awareness

Launch the product on a small scale, connect with shoe/footwear giants and look to expand the product's functionality and make more iterations.

FootSense

Lab-grade insight. Maker-grade budget. Blood. Sweat. Tears. A LOT of work. Even more help from mentors.

Questions?



People We Want to Thank

The amazing teammates we had here.

The ENTIRE Maker Team:

Dakshat Sir

Anool Sir

Vaibhav Sir

Miss Amber

Miss Sushmita

Sahil Sir

Miss Richa

Sankalp

Kinshuk

Sovam

Gunjan

Simran

Adishi

Shyam

Bhoomika

Arjun

Jastin / Kalki 01

Sebi

All Our friends here, and Makers Asylum

THANK YOU!!

for watching and always supporting us